








Time Series Analysis

Seminar 11: Foundation models for time series



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CPU**: Central Processing Unit
- **CRPS**: Continuous Ranked Probability Score
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **ETS**: Error, Trend, Seasonality (exponential smoothing state space models)
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **GPU**: Graphics Processing Unit
- **HICP**: Harmonised Index of Consumer Prices
- **MAE**: Mean Absolute Error
- **MASE**: Mean Absolute Scaled Error
- **US**: United States
- **WQL**: Weighted Quantile Loss

Today's Question and Seminar Structure

- The question of the seminar
  - **is a forecast made by a model that never saw the series better than a simple benchmark, and how do we check it?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the scores of the A exercises, computed by hand
  - a table and a figure for each B exercise
  - the corrected AI answer of C1
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - no model is run again: the notebook reads the saved forecasts of Lecture 11
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	pinball losses at one hour of load; MASE (mean absolute scaled error) of three forecasts of industrial production	Solved, Proposed	A1
A3, A4	the weighted quantile loss (WQL) of four origins; is an 80% interval calibrated?	Solved, Proposed	A3
B1, B2	zero-shot models on Romanian inflation; error by horizon for US retail sales	Solved, Proposed	B1
B3	is Chronos-2 significantly better? Diebold–Mariano tests on seven series	Solved	B3
C1	what is wrong in an AI answer about foundation models?	Proposed	A1–B3

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency, horizon	Forecast origins
electricity load, Romania	ENTSO-E	hourly, 48 hours	2025–2026, weekly
industrial production, Romania	Eurostat	monthly, 12 months	2016–2025
real GDP, Romania	Eurostat	quarterly, 4 quarters	2012–2025
HICP inflation, unemployment, Romania	Eurostat	monthly, 12 months	2016–2025
EUR/RON	BNR	daily, 20 days	2016–2026
retail sales, US	FRED (RSXFSN)	monthly, 12 months	2016–2025

- Forecasts of Lecture 11, saved once: seasonal naive, ETS, ARIMA and three open foundation models (Chronos-Bolt tiny and small, Chronos-2), run zero-shot on a CPU (central processing unit) Ansari et al. (2024)
- In the notebook: `backtest_rows`, `abs_errors`, `pinball`, `summarise`, `dm_test`; no account, key or GPU is needed

Prerequisites for Today (1/5): zero-shot forecasts and quantiles

- **Foundation model**: a network pretrained once on many series; **zero-shot**: it forecasts a new series from its recent values only, with no training on that series
  - Chronos divides the context $x_1, \ldots, x_C$ by its scale $s = \frac{1}{C} \sum_t |x_t|$, then maps x_t/s to one of 4093 bins (tokens) Ansari et al. (2024)
- **Output**: for each step $j = 1, \ldots, h$, nine quantiles $\hat{q}_\tau$, $\tau = 0.1, 0.2, \ldots, 0.9$
  - the median $\hat{q}_{0.5}$ is the point forecast; $[\hat{q}_{0.1}, \hat{q}_{0.9}]$ is the central 80% interval
  - the statistical baselines give Normal quantiles: $\hat{q}_\tau = \hat{y} + z_\tau \hat{\sigma}$, with $z_{0.9} = 1.2816$
- **Seasonal naive** with period m: $\hat{y}_{T+j} = y_{T+j-m}$ (the same hour last week, the same month last year)

Prerequisites for Today (2/5): the pinball loss and the coverage

- **Pinball (quantile) loss** of a quantile forecast q at level τ when y is observed

$$L_\tau(y, q) = \begin{cases} \tau\,(y - q), & y \geq q \\ (1 - \tau)\,(q - y), & y < q \end{cases}$$

  - for $\tau = 0.9$ an outcome above the quantile costs 9 times more than one below it: the loss is smallest, on average, at the true 90% quantile
  - $L_{\mathbf{0.5}} = \frac{\mathbf{1}}{\mathbf{2}}|y - q|$: half the absolute error of the median
- **Coverage** of the 80% interval: the share of outcomes y inside $[\hat{q}_{\mathbf{0.1}}, \hat{q}_{\mathbf{0.9}}]$; a calibrated interval covers about 80%
  - below 80%: too narrow (overconfident); above 80%: too wide

Prerequisites for Today (3/5): the weighted quantile loss (WQL)

- **WQL** of one forecast origin: all nine levels and all h steps, scaled by the size of the series

$$\text{WQL} = \frac{\text{num}}{\text{den}} = \frac{\sum_{\tau} \sum_{j=\mathbf{1}}^{h} 2L_{\tau}(y_{T+j}, \hat{q}_{\tau,T+j})}{9 \sum_{j=\mathbf{1}}^{h} |y_{T+j}|}$$

  - a discrete approximation of the CRPS of the whole predictive distribution Gneiting and Raftery (2007); smaller is better
- **Over many origins**: the ratio of the sums, $\sum_i \text{num}_i / \sum_i \text{den}_i$, not the mean of the ratios
  - origins with a large series (more load in winter) weigh more
- **Relative WQL**: the WQL of a model divided by the WQL of the seasonal naive on the same origins; below 1: better than the benchmark

Prerequisites for Today (4/5): MASE and relative scores

- **MASE** (mean absolute scaled error) of one origin, with the median as point forecast Hyndman and Koehler (2006)

$$\text{MASE} = \frac{\frac{1}{h}\sum_{j=1}^{h} |y_{T+j} - \hat{q}_{0.5,T+j}|}{s_m}, \qquad s_m = \frac{1}{T-m}\sum_{t=m+1}^{T} |y_t - y_{t-m}|$$

  - s_m: the in-sample MAE (mean absolute error) of the seasonal naive forecast one step ahead, on the context only
- MASE < 1 compares with an in-sample one-step benchmark, not with the seasonal naive of the test period
  - for a fair comparison use the **relative MASE**: MASE of the model / MASE of the seasonal naive, on the same origins
- **Several series**: the geometric mean of the relative scores, $\exp\left(\frac{1}{K}\sum_k \ln r_k\right)$; a gain of 20% and a loss of 20% do not cancel in the arithmetic mean

Prerequisites for Today (5/5): rolling origins and the Diebold–Mariano test

- **Rolling origin**: at each origin T_i every model forecasts h steps from the data up to T_i; then the origin moves forward (by one month, one week, 20 days)
  - a score from one origin is noisy; the backtest averages over many origins Hewamalage et al. (2023)
- **Diebold–Mariano (DM) test** Diebold and Mariano (1995): is the mean loss differential zero?

$$d_i = L_i^A - L_i^B, \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}} \approx N(0,1)$$

  - $\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{k=1}^{\ell}(1 - \frac{k}{\ell+1})\hat{\gamma}_k$: Newey–West long-run variance; forecast windows overlap when the step between origins is below h, so $\ell = \lceil h/\mathrm{step} \rceil - 1$
  - DM < -1.96: model A significantly better at 5%; > 1.96: significantly worse

A1: pinball losses at one hour [Solved]

Task

- Romanian load, origin 22 June 2026, hour 24 (23:00); actual $y = 5.803$ GW.
- Chronos-Bolt small: $\hat{q}_{0.1}, \hat{q}_{0.5}, \hat{q}_{0.9} = 5.184, 5.476, 5.776$.
- Seasonal naive: $4.591, 5.388, 6.185$ (median = the same hour one week earlier).
- 1. Compute $L_{0.1}$, $L_{0.5}$, $L_{0.9}$ for both forecasts and their sums.
- 2. Is y inside each 80% interval? Which forecast scores better?
- Report: six losses, two sums, two verdicts.

Solution

- 1. Bolt: y is above all three quantiles, so $L_\tau = \tau(y - q)$: $0.1 \cdot 0.620 = 0.062$; $0.5 \cdot 0.327 = 0.163$; $0.9 \cdot 0.027 = 0.025$; sum 0.250
- Naive: $0.1 \cdot 1.212 = 0.121$; $0.5 \cdot 0.415 = 0.208$; $y < \hat{q}_{0.9}$: $0.1 \cdot 0.382 = 0.038$; sum 0.367
- 2. Bolt: $y > \hat{q}_{0.9}$, outside its interval (width 0.59 GW); naive: inside (width 1.59 GW)
- Bolt still has the smaller loss: a narrow interval that misses by a little beats a wide one; one hour proves nothing, so we average over many hours (A3)

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A2: MASE of three forecasts [Proposed]

Solution discussed in class

Task

- Romanian industrial production, origin August 2025, $h = 12$ months. Absolute errors $|y_{T+j} - \hat{q}_{0.5,T+j}|$ (index points). Model: A1.

j	1	2	3	4	5	6	7	8	9	10	11	12
Chronos-2	3.96	0.56	0.49	3.49	0.23	7.23	4.16	3.17	5.10	4.93	3.88	4.19
Seas. naive	2.70	2.50	0.40	2.80	1.30	3.00	1.70	0.70	2.60	7.40	3.30	6.40

- 1. Compute the MAE of Chronos-2 and of the seasonal naive forecast.
- 2. The context (307 months) gives $s_{12} = 4.354$. Compute both MASE values and the relative MASE of Chronos-2; ETS has MASE 1.204.
- 3. Over all 116 origins the mean MASE is 1.017 for Chronos-2 and 1.252 for the seasonal naive. What does one origin say?
- Report: two MAE, three MASE, two ratios, one sentence.

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A3: the weighted quantile loss of four origins [Solved]

Task

- Romanian load, the last four weekly origins of 2026 (day.month); numerator and denominator of the WQL (48 hours, 9 levels); C2 = Chronos-2, N = seasonal naive:

Origin	num C2	num N	den	WQL C2	WQL N
08.06	52.4	198.9	2264	0.0232	0.0878
15.06	52.5	86.7	2233	0.0235	0.0388
22.06	142.0	219.6	2526	0.0562	0.0869
29.06	130.8	197.1	2798	0.0468	0.0705

- 1. Compute the WQL of each origin (the last two columns).
- 2. Compute the WQL of the four origins as the ratio of the sums, and the mean of the four ratios.
- 3. Compute the relative WQL of Chronos-2 (C2).

Solution

- 1. For example 377.7 is the sum of the C2 numerators; each WQL is num/den (table)
- 2. C2: $377.7/9821 = 0.0385$; mean of ratios 0.0374. Naive: $702.3/9821 = 0.0715$; mean 0.0710
- 3. Relative WQL $= 0.0385/0.0715 = 0.538$ (with the means of ratios: 0.527)
- the two ways differ when the denominators differ: the ratio of sums gives the high-load weeks more weight

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A4: is an 80% interval calibrated? [Proposed]

Solution discussed in class

Task

- Romanian load, 52 weekly origins $\times$ 48 hours $=$ 2496 hourly outcomes. Chronos-2: 1830 inside its 80% interval; ETS: 2432. Model: A3.
- 1. Compute the two coverages and the binomial statistic $z = (\hat{c} - 0.8)/\sqrt{0.8 \cdot 0.2/n}$, as if all hours were independent.
- 2. The coverage of each origin has standard deviation 0.197 for Chronos-2 (0.050 for ETS) across the 52 origins. Compute $t = (\hat{c} - 0.8)/(\mathrm{SD}/\sqrt{52})$.
- 3. Why is $|t|$ much smaller than $|z|$? Is each interval too narrow or too wide?
- Report: two coverages, four statistics, two verdicts.

B1: zero-shot forecasts of Romanian inflation [Solved]

- **Question**: do the foundation models beat the statistical baselines on Romanian monthly inflation, and did the 2021–2023 inflation surge change the ranking?
- **Inputs**: monthly HICP inflation, 12 months ahead, 117 origins from January 2016 to September 2025 (saved backtest)
- **Tasks**
  1. From the saved rows, compute for each model the mean MASE, the WQL (ratio of sums), the coverage and the ratios to the seasonal naive.
  2. Repeat the relative WQL for the origins before and after January 2021.
  3. Plot the relative WQL over rolling windows of 24 origins.
  4. Interpretation: is the gain of the foundation models stable over time?
- **Report**: a table of six models, eight ratios, one figure, one sentence
- Notebook: section B1

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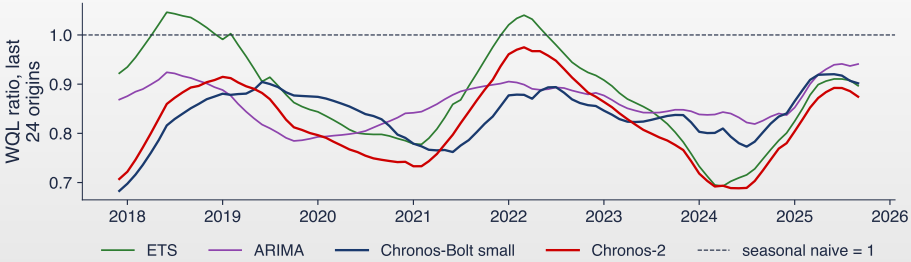
B1: solution [Solved]

Model	MASE	MASE ratio	WQL	WQL ratio	coverage 80%
seasonal naive	0.873	1.000	0.667	1.000	0.900
ETS	0.808	0.926	0.589	0.883	0.836
ARIMA	0.780	0.893	0.581	0.870	0.873
Chronos-Bolt tiny	0.720	0.825	0.537	0.806	0.858
Chronos-Bolt small	0.744	0.852	0.549	0.824	0.846
Chronos-2	0.726	0.831	0.542	0.812	0.800

- WQL ratio before / after January 2021 (60 and 57 origins): Chronos-2 0.761 / 0.851, Bolt small 0.772 / 0.862, ETS 0.884 / 0.882, ARIMA 0.851 / 0.885
- Interpretation: all models beat the seasonal naive and the foundation models are ahead on average; the gain shrinks after 2021 (in the window ending March 2022: 0.975) and the ranking changes over time

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B1: relative WQL over time [Solved]



- Chronos-2 is the best of the four models in 46% of the windows; its ratio ranges from 0.688 (window ending June 2024) to 0.975 (March 2022)
- In 2022 the inflation shock was new to every model: ETS fell behind the seasonal naive, the foundation models lost most of their lead

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B2: error by horizon for US retail sales [Proposed]

- **Question**: at which horizons does a zero-shot model help most, and does the size of the model matter?
- **Inputs**: US retail sales (FRED RSXFSN, billion USD), 12 months ahead, 117 origins, 2016–2025; model: B1
- **Tasks**
  1. From the saved absolute errors, compute the MAE of each model for each step $j = 1, \ldots, 12$.
  2. Divide by the MAE of the seasonal naive forecast at the same step and plot the ratios.
  3. Compare the three foundation models (tiny, small, Chronos-2) with ETS and ARIMA at $j = 1$ and $j = 12$.
  4. Interpretation: why do all ratios rise towards 1 at long horizons?
- **Report**: ten ratios, one figure, one sentence
- Notebook: section B2

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B3: is Chronos-2 significantly better? [Solved]

- **Question**: on which of the seven series is the gain of Chronos-2 larger than the noise of the backtest?
- **Inputs**: the WQL of each origin (num/den) of Chronos-2, ARIMA and the seasonal naive, seven series
- **Tasks**
  1. For each series compute $d_i = \mathrm{WQL}_i^{\mathbf{C2}} - \mathrm{WQL}_i^{\mathbf{rival}}$ for the two rivals.
  2. Compute the DM statistic with the Newey–West variance and $\ell = \lceil h/\mathrm{step} \rceil - 1$ lags.
  3. Count the series where Chronos-2 is significantly better or worse at 5%.
  4. Interpretation: does a better average WQL always mean a significant gain?
- **Report**: fourteen DM statistics, two counts, one sentence
- Notebook: section B3

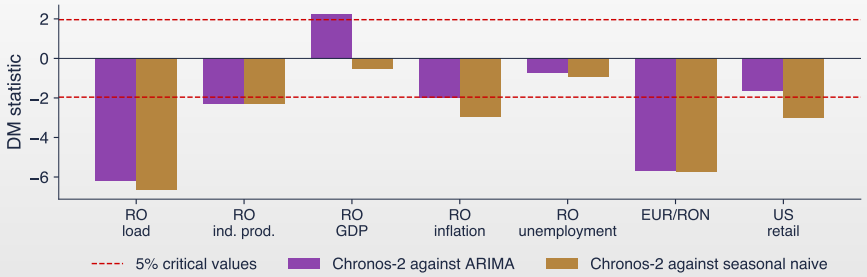
B3: solution [Solved]

Series	ℓ	DM vs ARIMA	p	DM vs naive	p
RO electricity load	0	−6.20	<0.001	−6.67	<0.001
RO industrial production	11	−2.32	0.020	−2.30	0.022
RO real GDP	3	2.25	0.025	−0.54	0.586
RO inflation	11	−2.00	0.045	−2.97	0.003
RO unemployment	11	−0.72	0.472	−0.95	0.344
EUR/RON	0	−5.71	<0.001	−5.72	<0.001
US retail sales	11	−1.65	0.099	−3.01	0.003

- Significantly better than ARIMA on 4 series, worse on 1 (RO GDP, DM $= 2.25$); better than the seasonal naive on 5 series, worse on none
- Interpretation: no; on unemployment the mean WQL is lower but DM $= -0.72$; overlapping 12-month windows leave few independent comparisons

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B3: the DM statistics in a chart [Solved]



- Bars below -1.96: Chronos-2 significantly better; above 1.96: significantly worse; the load gives the strongest evidence (DM $= -6.20$ against ARIMA)
- On quarterly GDP a well-specified ARIMA wins: Chronos-2 is better at only 36% of the origins

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C1: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise the backtest of Lecture 11. The answer:
- (a) Chronos-2 beats every baseline on all seven series, so ARIMA is obsolete.
- (b) The WQL over several origins is the mean of the WQL of each origin.
- (c) An 80% interval that covers 73% of the hours is well calibrated: 73 is close to 80.
- (d) MASE below 1 means the model beats the seasonal naive forecast on the test period.
- (e) Zero-shot means the model never saw these series, so the backtest cannot be contaminated.
- (f) Mean scaling lets one model forecast EUR/RON (about 5) and an index (about 100) with the same tokens.
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (A1–B3, notebook section C1).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- Pinball loss: $\tau(y - q)$ above the quantile, $(1 - \tau)(q - y)$ below; WQL = ratio of sums over levels, steps and origins
- MASE is scaled by an in-sample error: compare models with the relative MASE on the same origins
- Coverage tests must respect the dependence of the hours of one origin
- Chronos-2 is the best model on average, but not on every series, and not always significantly
- An AI summary of a benchmark is a draft: check the aggregation, the benchmark and the test

After the seminar

- Lecture 11 gives the details
  - tokens and patches, the architecture of Chronos, the full backtest, context length, contamination
- Try the [Proposed] tasks in the notebook
- Project idea: add a new Romanian series and repeat B1 and B3 with the saved functions
- Reading: Ansari et al. (2024); Gneiting and Raftery (2007); Hyndman and Koehler (2006); Diebold and Mariano (1995)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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